

Unveiling Structural Distributional Bias in Collaborative Filtering for Ultra-Sparse Datasets

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Abstract— Collaborative Filtering is one of the most widely used rating prediction methods for recommender systems in recent years. It is based on a solid and verified rationale, it supports explainability and usually delivers satisfactory results. However, its performance declines when applied to ultra-sparse datasets. Despite extensive research, the root cause of its failure in such environments still requires further investigation. This work systematically investigates structural distribution biases in CF ultra-sparse datasets, by examining 11 datasets in this category. The findings of this study are threefold. First, this study reveals that the rating distribution within ultra-sparse datasets, exhibits a strong skew toward boundary ratings, thus creating a non-normal rating structure that challenges the traditional Collaborative Filtering algorithms. Secondly, through extensive experiments, employing four different similarity metrics and using three nearest neighbor selection thresholds, it asserts that ratings with values that are underrepresented in the dataset tend to produce significantly higher error margins — up to four times larger than predictions corresponding to the most frequent rating values. The third and most important finding of this study is the phenomenon of mean rating attraction. Specifically, it reveals that the rating predictions are consistently anchored to user and item averages, a process which essentially computationally smooths the rating predictions, instead of recognizing true and unique non-conforming preferences. The above conclusions were found to be independent of both the selection of user similarity metric and the neighbor selection threshold; rather, they are grounded on the inherent structural bias of the input data. This work offers a new perspective for Collaborative Filtering algorithm design and implementation, by identifying structural bias as an important dimension that needs to be handled by rating prediction algorithms, along with the dimension of similarity metrics between users and items.

Keywords—*Collaborative Filtering, Recommender Systems, Ultra-Sparse datasets, Structural Distributional Bias, Similarity Metrics, Performance Evaluation*

I. INTRODUCTION

Recommender Systems (RecSys) play a significant role in every user's day-to-day online experience because of the massive quantity of online information, and thus the vast options users are faced with. By providing consumers with filtered views of available selections in the form of personalized product offerings, based on their individualized preferences, RecSys have become ubiquitous throughout E-

commerce, Streaming Services, Social Media Platforms and many other industries [1], [2]. Collaborative Filtering (CF) plays a dominant role in RecSys, relying on the fundamental premise that if two users have exhibited alike behaviors in the past, they are likely to do so again. While Content-Based RecSys require detailed analyses of both user and item meta-data to function effectively, CF uses only historical user-item interactions as input [3], [4].

Traditionally, the performance of a CF algorithm is determined by how precisely it predicts a user's likings for items that the user has not seen. In algorithm evaluation, two statistical metrics commonly used to measure the degree of accuracy of these prediction capabilities are Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) [5]. Although this represents the initial step in the recommendation pipeline —predicting unknown user ratings— the transition from those predictions to actual recommended products can mask a serious shortcoming: the assumption that lower MAE/RMSE values are indicative of higher quality, and more relevant recommendations. This paper argues that this assumption is frequently incorrect, especially when dealing with very sparsely populated user-item matrices.

Although CF-based recommendation engines are highly successful, their performance is known to be hampered under conditions of high data sparsity. With respect to most real-world applications, the user-item interaction matrix is very sparse. For example, although there may be approximately 5000-7000 movie/series titles contained in the entire Netflix catalog; the typical user has interacted with fewer than 100-200 of those items. As such, in ultra-sparsely populated environments, where interaction density is generally less than 0.1% or even significantly smaller, traditional similarity measures (i.e., Pearson Correlation and Cosine Similarity) fail [6], [7], [8]. Due to the lack of sufficient ratings among common groups of users, similarity scores calculated using the aforementioned similarity metrics become unstable, resulting in either completely arbitrary results (i.e., random), or overly stereotypical results (c.f. Fig. 1).

While existing literature focuses primarily on refining similarity metrics to mitigate sparsity, the present work shifts this perspective [9], [10], [11]. We argue that the fundamental challenge lies not merely in data volume, but in the inherent

structure of the ratings — a phenomenon we term *Structural Distributional Bias*. Through an empirical analysis of 11 diverse ultra-sparse datasets, widely used in the field of CF RecSys, we demonstrate that the rating distribution within the datasets is neither uniform nor Gaussian. Instead, ratings tend to cluster around extreme values, thereby creating significant asymmetries.

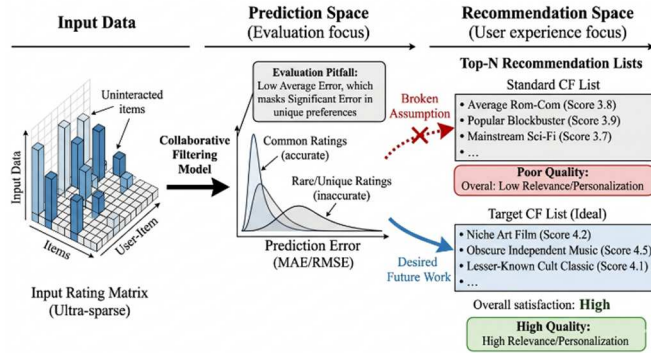


Fig. 1. Prediction-recommendation gap due to structural bias in CF

We find that traditional CF models fail to accommodate this structural skew, as their predictions exhibit a systematic ‘gravity bias’ toward the user and item averages ($avgU$ and $avgI$, respectively). In essence, the CF system does not predict actual preferences, but ‘anchors’ itself to the statistical means of the data, effectively suppressing user traits that deviate from typical cases.

This paper contributes to the research community in three primary ways:

1. **Diagnostic Analysis:** We identify Structural Distributional Bias, demonstrating that rating distribution is a fundamental structural feature that dominates the distribution of prediction errors, regardless of both the similarity metrics and the nearest-neighbor selection configuration
2. **Performance Dissection:** We analyze MAE and RMSE across rating scales (1/5 to 5/5), demonstrating that CF accuracy is asymmetric and strongly correlated with the frequency of each rating category.
3. **The Anchoring Effect:** We reveal the *statistical anchoring phenomenon*, where predictions are primarily driven by the $avgU$ and $avgI$ factors, effectively hampering the CF system’s ability to identify the user’s personalized preferences.

Through this analysis, we aim to demonstrate that improving RecSys performance on ultra-sparse data requires a shift from merely optimizing similarity metrics, to successfully coping with the inherent structural bias in input data distributions.

The rest of this paper is organized as follows: Section II discusses the relevant literature, while Section III briefly presents the foundations of the present work, to ensure self-containment. Section IV presents the empirical findings and analysis, and provides a discussion, and finally Section V concludes the paper and outlines directions for future work.

II. RELATED WORK

CF continues to serve as one of the primary building blocks of RecSys, using past user interaction history to help

estimate potential new interests. In recent years, most of the research has been dedicated to enhancing the prediction accuracy of CF, particularly when faced with very sparsely populated datasets. Many contemporary studies have looked at alternative similarity metrics, novel ways to select neighbors, and numerous optimizations to CF algorithms. Furthermore, recent studies also show that many small changes to parameters can greatly affect the overall performance of the system.

To overcome the shortcomings of traditional similarity measures in extremely sparse scenarios, several researchers have recently attempted to utilize auxiliary information in addition to exclusively co-rated items. For example, Guan et al. [7] introduced a robust hybrid similarity model based on the Wasserstein metric to circumvent the issues associated with metrics like KL divergence. Their similarity model introduces heuristics designed specifically to limit the distortion caused by popular items, which frequently dominate the similarity in sparse datasets; consequently, their approach prioritizes actual interactions over popular ones. Additionally, Okkalioglu [12] proposed a multi-criteria method that uses alpha-divergence to combine local and global similarities. By employing the probability density distribution of ratings as a surrogate measure of similarity, the multi-criteria method provides accurate estimates of similarity when there is little or no overlap between users; thereby establishing a solid foundation for estimating similarities in fragmented interaction matrices. Similarly, the UDIS method by Koochi et al. [13] focused on fusing multiple data sources. Through SVD clustering, UDIS combines predictions derived from user-based, item-based, demographic-based and semantic-based sources; thus, it increases the density of the predictive signal by combining predictions from multiple data sources. Furthermore, because UDIS is an ensemble method, it demonstrates that when no user-item interaction data exist, developing auxiliary knowledge is important for ensuring good-quality recommendations.

Interestingly, although much of the development effort in recent years shifts towards deep learning architectures, recent comparative studies show that memory-based CF remain relevant today. Specifically, Shetty et al. [10] demonstrated that well-tuned versions of memory-based algorithms (e.g., SVD++) can achieve comparable levels of accuracy (or even better) compared to state-of-the-art deep learning architectures when properly configured.

In conclusion, these developments collectively highlight that the accuracy bottleneck does not relate to the use of deep architecture; rather it relates to the ability to extract useful information from the available data.

However, a critical shortcoming remains: whether through improved similarity estimation, integration/fusion of multiple signals, or fine-grained tuning of hyperparameters, ultimately all prior studies aim to minimize error metrics (RMSE and/or MAE). However, due to this focus on minimizing error metrics alone, previous studies neglect the fundamental structural bias in the rating distributions themselves — a ‘gravity’ effect that drives predictions toward the mean regardless of the complexity of the employed model.

Concurrently, more recent studies begin to question the appropriateness of relying solely on error metrics as indicators of user satisfaction with recommended content. For instance, González et al. [14] demonstrated that biases inherent in CF

typically arise from sociopsychological aspects of the user interaction data rather than from properties of the models used to generate those predictions. Therefore, González et al. proposed ‘soft matrix factorization’ (SoftMF) — a method to address these imbalances inherent in the user interaction data to improve the quality of generated recommendations; thereby emphasizing that improvement in recommendation quality necessitates an examination of these inherent imbalances rather than simply increasing architectural complexity. Expanding upon this conceptually similar idea, Zhu et al. [15] identified ‘popularity-opportunity bias’ — an inherent characteristic of matrix factorization— whereby items that are popular receive a higher rank than otherwise equivalent preferred items of other users. These authors highlighted that reliance on error-minimization produces a feedback loop wherein unique items are systematically under-represented; and therefore, moving beyond reliance on standard error metrics is required to avoid this systematic underrepresentation.

Shao et al. [16] addressed limitations related to fair treatment based solely on user identities by arguing that sensitive information about individual users is frequently encoded through item correlations in CF systems. To address this issue, they developed FairCF — a framework that incorporates fairness within the embedding space via both user and item classifiers; thereby demonstrating that unbiased representation requires a holistic approach incorporating both entity-specific and collaborative representations.

Borges & Stefanidis [17] investigated ‘feature-blind’ unfairness —i.e., bias present in CF systems even without explicit sensitive attributes— and employed a Variational Autoencoders (VAE)-based framework to develop a fairness term incorporated into the loss function; resulting in a 42% reduction in positional bias at a cost of only a 26% reduction in recall. Borges & Stefanidis provided evidence supporting the idea that equitable results can be achieved even in extremely data-sparse environments.

Overall, these works established that existing evaluation frameworks do not possess sufficient detail to determine structural biases inherently determining long-term performance. Focusing exclusively on error minimization leads these approaches to ignore how rating distributions cause predicted values to be anchored near some mean value; thereby establishing a foundation for more detailed evaluation criteria.

Finally, biases in CF have been studied from multiple angles including static popularity bias and long-term exposure effects. Mansoury et al. [18] studied this phenomenon dynamically through modeling a feedback loop, wherein system-generated recommendations create future user behavior that is subsequently absorbed back into the training dataset. Their empirical results showed that this feedback loop dramatically amplified popularity bias -- producing dramatic declines in overall diversity and creating homogeneous user preference profiles; with disproportionate negative impacts felt by minority user subgroups. Extending these feedback mechanisms, Xu et al. [19] modeled the dynamical aspects of these interactions using causal graphs containing cyclic loops. The authors empirically showed that if left unchecked, these loops will result in ‘echo chambers’, causing users to become locked into increasingly narrow content exposures. To break this determinism cycle of algorithmic reinforcement the authors proposed partialCCF —a Dynamic Causal CF model

that utilizes causal reasoning and counterfactual adjustments to prevent this type of determinism.

Building on this conceptual expansion, Bartley et al. [20] extended the scope of this perspective by investigating how personalization influences user perceptions through pre-existing social networks; showing that less complex feed-based algorithms amplify “exposure bias” more than advanced deep-learning based algorithms; pointing to “effective degree-attribute correlation” as a key mechanism to reduce exposure-bias.

Bellina et al. [21] formally analyzed recommendation-driven polarization using methodologies from statistical physics; showing that polarization occurs at specific system phases — disorder, consensus and polarization — and that by adjusting weights for bias — it is theoretically possible to construct algorithms capable of providing personalized recommendations while reducing systemic fragmentation.

While all the previously discussed studies —including recent debiasing or fairness-aware CF frameworks— have made considerable advances in modeling the dynamic effects of bias (from forming echo chambers to phase transitions in polarization), none have examined how rating distributions impose fundamental structural constraints on algorithmic performance.

Therefore, we conclude that prior studies and contemporary fairness interventions have largely viewed failure and bias in RecSys as either an external interaction issue or an artifact stemming from feedback processes. In contrast, our work represents a paradigmatic shift — directing attention away from examining external issues or internal processes as causes for recommendation failure toward evaluating structural flaws inherent in the recommendation architecture itself. By deconstructing the mathematical inevitability of the anchoring effect and the asymmetric error scaling inherent in standard rating distributions, we provide the diagnostic framework necessary to understand why CF models struggle to move beyond global mean approximations.

III. FOUNDATIONS

In this Section, we briefly overview the fundamental concepts used in the present work, to ensure self-containment. Specifically, we outline the complete process of CF prediction formulation, which comprises (a) user similarity metrics, (b) the neighborhood selection strategy, (c) the rating prediction formulation, as well as (d) the rating prediction error-deviation metrics that are used to quantify and evaluate the performance of the rating predictions.

A. User Similarity Metrics

The first step towards producing CF rating predictions is computing the similarity between pairs of users. The most popular metrics utilized in CF research are Pearson correlation (PC) and Cosine similarity (CS) [22], [23], [24]. This study extends the domain of this research through the use of two new versions of the base metric, the Sigmoid Pearson (SP) and the Constrained Pearson (CP) [4], [25]. These additional versions will allow for increased granularity when studying how varying degrees of structural distributional biases impact the performance of these models within very sparse rating environments. Table I presents the definitions of the aforementioned four metrics employed in the present study.

B. Neighborhood Selection Strategy

Once the similarities between all pairs of users have been calculated, the neighborhood selection phase commences, employing either the K-Nearest Neighbors (KNNs) method or the similarity thresholding strategy (THR) [4], [26], [27]. In the present work, we utilize the THR strategy, as it imposes a more rigorous requirement on the selected NNs' quality. By defining a threshold T , we ensure that a neighbor n is included in the prediction process for user u (i.e., the user for whom the rating prediction is generated), only if the similarity between u and n surpasses this threshold.

TABLE I. CF USER SIMILARITY METRICS

Pearson	$PC(u_1, u_2) = \frac{\sum_{k \in CR_{u_1, u_2}} (r_{u_1, k} - \bar{r}_{u_1}) \cdot (r_{u_2, k} - \bar{r}_{u_2})}{\sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_1, k} - \bar{r}_{u_1})^2} \cdot \sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_2, k} - \bar{r}_{u_2})^2}}$
Cosine	$CS(u_1, u_2) = \frac{\sum_{k \in CR_{u_1, u_2}} r_{u_1, k} \cdot r_{u_2, k}}{\sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_1, k})^2} \cdot \sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_2, k})^2}}$
Constrained Pearson	$CP(u_1, u_2) = \frac{\sum_{k \in CR_{u_1, u_2}} (r_{u_1, k} - r_{med}) \cdot (r_{u_2, k} - r_{med})}{\sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_1, k} - r_{med})^2} \cdot \sqrt{\sum_{k \in CR_{u_1, u_2}} (r_{u_2, k} - r_{med})^2}}$
Sigmoid Pearson	$SP(u_1, u_2) = PC(u_1, u_2) \cdot \frac{1}{1 + e^{\frac{- CR_{u_1, u_2} }{2}}}$
$I(u_a)$: represents the set of items that user u_a has rated $CR_{a,b}$: represents the set of items evaluated by both users u_a and u_b , i.e. $CR_{a,b} = I(u_a) \cap I(u_b)$ $r_{u,k}$: represents the rating assigned to item k by user u . $\bar{r}_u$: represents the average value of all ratings given by user u r_{med} : represents the median value of the rating scale.	

To capture a comprehensive view of the CF system's behavior, in this work we consider three distinct thresholds: $T \in \{0.0, 0.25, 0.5\}$. $T = 0.0$ is used as a baseline to allow a large number of neighbors to be included, while $T = 0.5$ limits the representation to only users with a high degree of likeness to the current user, thus providing insight into the appearance of structural biases based on limited choices.

This methodology provides robustness regarding the analysis of structural distributional bias, as it allows the neighborhood size to vary naturally according to the dataset's density and the actual similarities observed in the dataset, rather than forcing the inclusion of potentially dissimilar neighbors — a common limitation inherent in the KNN approach.

C. Rating Prediction Formulation

Lastly, rating predictions are generated using a formulation that aggregates the ratings given by the user's NNs for a specific item. The similarity of each NN to the user, as computed in the previous subsection, acts as a weighting factor for the NN's rating. Previous research has experimentally demonstrated that in very sparse environments optimal results are obtained using the mean-centered formula (MCF) [4], [8]. MCF accounts for individual user rating biases by subtracting the average rating of each NN and its definition is given in Equation (1). In Equation (1), the similarity between the user and each NN is denoted as “sim(u_1, u_2)”.

$$MCF_{u_1, i} = \bar{r}_{u_1} + \frac{\sum_{u_2 \in NN_{u_1}} \text{sim}(u_1, u_2) \cdot (r_{u_2, i} - \bar{r}_{u_2})}{\sum_{u_2 \in NN_{u_1}} \text{sim}(u_1, u_2)} \quad (1)$$

Maintaining the definitions from Table I, $r_{u,k}$ represents the rating assigned to item k by user u , and $\bar{r}_u$ represents the average value of all ratings set by user u .

D. Performance Evaluation Metrics

The prediction accuracy of a CF system is primarily assessed by comparing the predicted ratings against the users' actual ratings, typically using error-based metrics like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

$$MAE = \frac{1}{N} \sum_i |prediction_i - rating_i| \quad (2)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_i (prediction_i - rating_i)^2} \quad (3)$$

The first metric calculates the average magnitude of prediction errors, while the latter squares the differences before averaging, thereby penalizing larger errors more heavily. Both these metrics are used in our work, and their definitions are given in Equations (2) and (3), respectively.

IV. EXPERIMENTAL EVALUATION

This study investigates structural distributional bias across ultra-sparse datasets, highlighting systemic limitations in CF. The purpose of this Section is to quantitatively evaluate how significantly skewed and boundary-dominant rating distributions undermine traditional predictive assumptions and to identify the sources of performance decline.

A. Dataset Characteristics and Experimental Setup

To evaluate the effect of structural distributional bias, we use 11 real-world datasets, summarized in Table II. These datasets are widely utilized in CF research to benchmark system performance in sparse environments. Additionally, the choice of these datasets enhances the generalizability of our results, since the chosen datasets:

- encompass a broad spectrum of domains (including movies, music, video games, software, and books), thereby eliminating any potential domain-specific bias.
- vary significantly in both size (i.e., number of ratings) and rating density, ranging from 0.24% (Yahoo Movies), which is considered simply sparse, down to 0.005% (LibraryThing), which is considered ultra-sparse.

TABLE II. DATASET INFORMATION

Name	Unique Users	Unique Items	Total Ratings	Density %
Amazon Baby Products [28]	151K	36K	1.24M	0.02
Amazon CDs & Vinyl [28]	124K	89K	1.55M	0.01
Amazon Industrial & Scientific [28]	51K	26K	413K	0.03
Amazon Movies & TV [28]	657K	198K	7.44M	0.006
Amazon Software [28]	146K	18K	1.28M	0.05
Amazon Videogames [28]	95K	26K	815K	0.03
CiaoDVD [29]	2K	15K	36K	0.12
Epinions [30]	40K	140K	665K	0.01
LibraryThing [31]	71K	385K	1.39K	0.005
Yahoo Movies [32]	8K	12K	221K	0.24
Book-Crossing [33]	23K	319K	1.03M	0.01

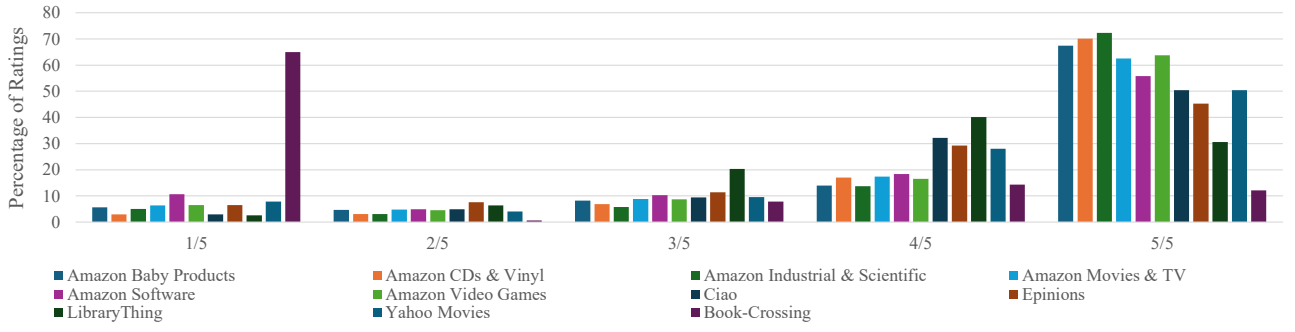


Fig. 2. Rating value distribution across the 11 examined datasets

The variety of domains, sizes and densities of the datasets used is very important because it allows us to demonstrate and verify that the observed bias is of a generic nature, and therefore independent of a particular domain or dataset.

In order to determine the effects of structural distributional bias on CF performance, we conduct three systematic experiments:

1. We perform a frequency analysis of all rating values across the 11 datasets to establish the baseline distributional characteristics of the input data. The goal of this analysis is to provide insight into the distributional properties of the input data, including the quantification of the degree of skewness of the data, and the identification of the degree of concentration of the ratings near the boundary values.
2. Using a Leave-One-Out cross validation method, we evaluate the predictive performance of the memory-based CF algorithm utilizing the MCF [34], [35]. Our testing includes a variety of factors including four methods for calculating similarities, and three thresholds concerning the formulation of NN neighborhoods (both described in detail in the previous Section). As part of evaluating system success in user rating prediction, we calculate the MAE and RMSE, both globally and for each distinct rating value from 1/5 through 5/5. A detailed evaluation of the predictive performance of the system is necessary in order to identify differences in performance over various levels of rating frequency.
3. We investigate the phenomenon of mean rating attraction by analyzing the underlying user and item signals. For every real rating, we calculate the average of the corresponding user and item averages and finally we group results by real rating category (1/5 to 5/5). This experiment is designed to reveal whether prediction errors correlate with the CF system's tendency to anchor estimates to these average values, thereby confirming the initial hypothesis concerning the presence of a structural bias.

B. Analysis of Prediction Performance and Distributional Bias

In this section, we present the results obtained from the experimental procedure described above. Our objective is to determine (a) how structural distributional bias manifests in the prediction accuracy of the CF models, and (b) the extent to which mean rating attraction affects the recovery of underrepresented user preferences.

First, we analyze the distributional characteristics of the 11 datasets. As illustrated in Fig. 2, the rating distribution is heavily polarized. All datasets exhibit a systematic deviation from the uniform distribution, with the majority of ratings being clustered at the bounds of the scale. Specifically, the 5/5 category is dominant for 10 out of the 11 datasets, whereas the Book-Crossing dataset shows a reverse trend, with a sharp frequency peak in the 1/5 category. In effect, we observe a structural imbalance, where in each dataset some categories are dominant, accounting for over 60% of the ratings. This imbalance will be further investigated concerning its effects on (a) on rating prediction accuracy and (b) the mean rating attraction phenomenon. Note that the Book-Crossing dataset has a range of [0, 10] and is mapped to the [1, 5] range using Equation (4).

$$mappedRating = \max(1, \left\lceil \frac{initialRating}{2} \right\rceil) \quad (4)$$

Fig. 3 (PC) and Fig. 4 (CS) provide empirical evidence of the 'inversion' effect. In the 10 Amazon-like datasets, the MAE starts with very high values (even > 2.0) in the low-frequency categories (1/5 and 2/5) and its value drops while moving towards the high-frequency categories (4/5 and 5/5). Conversely, in the Book-Crossing dataset (in which 1/5 ratings are the majority), MAE is minimized for the 1/5 and 2/5 rating categories and peaks at 5/5 ratings.



Fig. 3. Average MAE per rating category for the 10 Amazon-like datasets versus the Book-Crossing dataset, using the PC similarity metric at $T=0.0$

This behavior is found to be consistent across similarity metrics (including CP and SP), with identical trends being observed. This fact confirms that prediction errors are not a circumstantial phenomenon in a specific dataset, but a systemic consequence of distributional bias. CF models systematically exhibit errors when predicting preferences in minority categories, as the reliance on 'global views' drives predictions toward the majority sentiment, regardless of the metric employed.



Fig. 4. Average MAE per rating category for the 10 Amazon-like datasets versus the Book-Crossing dataset, using the CS similarity metric at $T=0.0$

To further validate that this bias is independent of the threshold parameter (T) tuning, Fig. 5 compares the category-specific MAE for the Pearson metric using a higher (compared to $T=0.0$ in Fig. 3) threshold of $T=0.5$. As shown in Fig. 5, the MAE distribution across categories remains virtually identical. This consistency confirms that distributional bias is a rigid structural characteristic of the CF process, rather than a side-effect of the chosen similarity threshold, as well.



Fig. 5. Average MAE per rating category for the 10 Amazon-like datasets versus the Book-Crossing dataset, using the PC similarity metric at $T=0.5$



Fig. 6. Average RMSE per rating category for the 10 Amazon-like datasets versus the Book-Crossing dataset, using the PC similarity metric at $T=0.0$

Lastly, to ensure that our findings are not dependent on the choice of error metric either, Fig. 6 presents category-specific performance using the RMSE metric when using the PC at $T=0.0$. The observed trends are identical to those of the respective analysis using the MAE metric (Fig. 3), since in both cases the model exhibits larger errors in minority classes and smaller errors in majority classes.

Recapitulating, the results of experiments with different similarity metrics, neighborhood formation thresholds, error metrics, and dataset domains affirm that this behavior is consistent across all other configurations, thus confirming

that the identified distributional bias is a fundamental and robust property of the underlying CF mechanism.

In Fig. 7, we present the mean user-centric rating (avgU) categorized by the ground truth rating ($\text{rating} \in \{1/5, 2/5, 3/5, 4/5, 5/5\}$). The data show an increasingly positive relationship with regard to the ratings of each user group as the ground-truth rating progresses from 1/5 to 5/5. This trend is consistent for all 10 datasets. For example, in CDs & Vinyl, the average ratings progress from 3.52 to 4.48 (3.52, 3.78, 4.00, 4.23, and 4.48); whereas in Yahoo Movies, a comparable increase is observed (3.34, 3.76, 3.91, 4.08, and 4.24). As such, this demonstrates the presence of the anchoring effect in the interaction data and how users can be statistically grouped by their tendency to rate higher or lower. This is attributed to the fact that when a user enters low (high) ratings, their mean rating tends to drop (rise); conversely users with low (high) average ratings are bound to have given low (high) ratings.

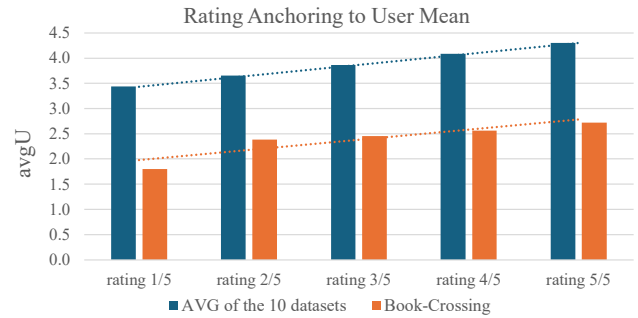


Fig. 7. Average rating distribution relative to avgU

As illustrated in Fig. 8, a symmetric structural dependency is observed regarding item-centric ratings (avgI). There also appears to be a consistent pattern where avgI increases monotonically with respect to the ground-truth value in this context. Therefore, it can be asserted that there exists a relationship between user ratings and item ratings (the two dimensions of the interaction matrix) such that each side consistently supports the other.

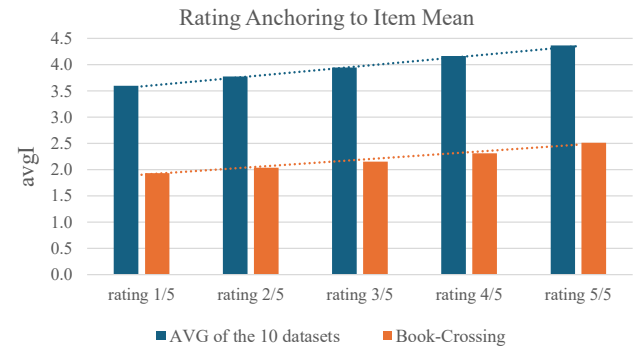


Fig. 8. Average rating distribution relative to avgI

Similarly, the domain independence of the phenomenon is demonstrated through examples, including the CDs & Vinyl dataset. In the CDs & Vinyl dataset, the avgI follows a specific order (e.g., 3.85, 4.06, 4.22, 4.36, 4.53), while the Yahoo Movies dataset demonstrates a similar order (i.e., 3.44, 3.61, 3.82, 4.06, 4.32). Overall, evidence is provided which suggests that the ‘anchoring effect’ is a common trait among interaction data topologies that relate to both users and items being statistically grouped based upon their respective patterns of rating.

C. Discussion

The findings from this study show that the performance of CF algorithms declines in ultra-sparse conditions, primarily due to structural biases inherent in the datasets, rather than being limited by suboptimal parameter choices. The primary driver of reduced CF performance is the highly skewed nature of the rating distributions, observed across all 11 tested datasets (illustrated in Fig. 2).

Beyond this high degree of skewness, there exists an inverse relationship between rating prediction errors (both MAE and RMSE) and rating frequency, as demonstrated in Figs. 3-6. Consequently, while the CF algorithm exhibits low errors when predicting ratings for frequent classes, it fails to capture extreme user preferences in infrequent ones, resulting in larger error magnitudes. This claim is further supported by the dependence of predicted ratings on user and item averages (as illustrated in Figs. 7 and 8).

Regardless of the dataset domain or algorithm configurations—including the similarity metric(s) used and neighborhood formation thresholds—the CF algorithm consistently estimates predictions as localized means and is, consequently, structurally tied to the typical behaviors of both users and items. Therefore, irrespective of the actual similarity (or dissimilarity) level between two users or two items, the use of traditional neighborhood-based CF, in ultra-sparse contexts, will formulate predictions that are subject to the inherent biases associated with the distribution of user-item interactions. Furthermore, our empirical evidence demonstrates that such models preferentially minimize overall error rates, rather than accurately reflecting real interaction diversity.

To bridge this gap and translate these insights into actionable algorithmic improvements, we outline three key design guidelines for next-generation CF frameworks operating under ultra-sparsity. First, to counter Boundary Polarization, data preprocessing pipelines should incorporate distribution-aware normalization techniques (e.g., quantile adjustments) to balance the skewed rating fields before similarity computations. Second, to mitigate Asymmetric Error Scaling, optimization functions must pivot toward cost-sensitive learning paradigms, assigning higher penalty weights to errors occurring within underrepresented minority classes. Finally, to bypass the Mean Rating Attraction, prediction formulation methods should integrate dynamic regularization or variance-enforcing penalty terms; this would prevent estimated scores from aggressively collapsing toward local user/item means when co-rated counts are critically low, thereby preserving unique individual preferences.

It is worth noting that while an extensive comparative evaluation against existing state-of-the-art debiasing algorithms falls outside the diagnostic scope of this paper, our findings establish the structural baseline necessary for such future benchmarks. Traditional algorithmic evaluation often overlooks how ultra-sparsity inherently constrains rating structures; thus, uncovering these mathematical limitations is a prerequisite before any subsequent mitigation framework can be effectively assessed.

V. CONCLUSIONS AND FUTURE WORK

In this work, we presented a comprehensive study aimed at unveiling the underlying structural distributional bias in CF, when dealing with ultra-sparse datasets.

Through an empirical analysis of 11 diverse ultra-sparse datasets, widely used in the field of CF RecSys, we demonstrated that the rating distribution within the datasets does not follow either the uniform or the Gaussian distribution; rather, users tend to exhibit extreme rating preferences, thereby creating significant skewness in the user-item rating matrix.

The experimental evaluation presented in Section IV confirmed the three core hypotheses examined in this work:

1. **Diagnostic Analysis:** We identified Structural Distributional Bias, demonstrating that rating distribution is a fundamental structural feature that dominates the distribution of prediction errors, regardless of the similarity metrics or the NN selection configuration.
2. **Performance Dissection:** We analyzed MAE and RMSE across rating scales (1/5 to 5/5), demonstrating that CF accuracy is asymmetric and strongly (inversely) correlated with the frequency of each rating category.
3. **The Anchoring Effect:** We revealed the ‘statistical anchoring phenomenon’, where predictions are actually primarily driven by the avgU and avgI factors, effectively hampering the CF system’s ability to identify the personalized user preferences.

These findings prove that the anchoring of ratings to user and item averages is a fundamental limitation of the CF paradigm in ultra-sparse domains.

In our future research, we plan to focus on developing debiased CF architectures that explicitly implement the proposed distribution-aware and variance-enforcing design guidelines, thereby avoiding the anchoring effect through identifying and prioritizing user-item interactions that more accurately convey actual user preferences. Furthermore, the experimental results suggest that CF algorithms operating on ultra-sparse datasets tend to formulate predictions that follow the user and item rating histories, without reflecting the actual user preferences towards items. Effectively, this behavior intensifies the popularity bias problem [36] and the filter bubble phenomenon [37] resulting in loss of serendipity and diminished user experience.

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